

# Relative Velocity: Crossing a Flowing River

Relative Velocity: Boat Crossing a River · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

## NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://ophysics.com/k11.html>

Curriculum links: motion

## Learning intentions

- I can add velocity vectors to find a resultant velocity relative to the ground.
- I can distinguish a boat's velocity relative to the water from its velocity relative to the ground.
- I can calculate the heading needed to cross a river directly, compensating for the current.

## Getting started

Open the simulation and set the boat's speed (relative to the water), its heading, and the speed of the river current. Run the simulation and observe the boat's actual path.

## Tasks

1. Point the boat directly across the river (perpendicular to the banks) with a moderate current running. Sketch a vector diagram predicting the boat's path over the ground, then run the simulation to check.

*Working:*

2. Using your boat speed and the current speed from Task 1, calculate the boat's resultant speed over the ground and the angle it drifts downstream.

*Working:*

3. Adjust the boat's heading (angling it upstream) until it crosses directly to a point straight across from its start. Calculate the heading angle needed, then compare it to what you found by trial and adjustment.

*Working:*

4. Repeat Task 3 for three different current speeds, recording the heading angle needed and the resulting crossing time for each.

**Current speed**

**Heading angle needed**

**Crossing time**

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## Extension (optional)

Explain how a pilot flying in a crosswind faces the same problem as the boat in this simulation, and why

they need to point the aircraft's nose somewhat into the wind to fly a straight path over the ground.

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